MIL-C-46344A(MU)
14 August 1970
SUPERSEDING
MIL-C-46344(MU)
30 July 1962

MILITARY SPECIFICATION

COMPUTERS, BALLISTICS: M13A2 AND M13A1DE1

1. SCOPE

1.1 This specification covers two types of instruments known as Ballistics Computer M13A2 and Ballistics Computer M13A1DE1, used to receive the log of range data from a range finder calibrated for range measurement in meters and compute superelevation for various ammunitions and transmit same to a gun sighting system.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture, and Inspection of
MIL-P-14232	Parts Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F8621000	Computer, Ballistics: M13A2
F10555194	Computer, Ballistics: M13A1DE1

FSC 1220

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PACKAGING DATA SHEETS

8621000	Packaging of Computer, Ballistics: M13A2
10555194	Packaging of Computer, Ballistics: M13A1DE1

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The computers shall be manufactured in accordance with Drawing F8621000 “Ballistics Computer, M13A2" and Drawing F10555194 “Ballistics Computer, M13A1DE1” and drawings pertaining thereto.

3.3 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Part identification and marking.
(e) Electrical.
(f) Workmanship.

3.4 Environmental.

3.4.1 Storage temperatures.- The computer shall show no evidence of physical failure and shall meet the performance requirements of 3,4.2.1 at standard ambient temperature after having been exposed and thermally stabilized at ambient temperatures of plus 160 and minus 80 degrees Fahrenheit.

3.4.2 Extreme temperaturre operability.- When exposed and thermally stabilized at ambient temperatures of plus 150 and minus 40 degrees Fahrenheit the computer shall meet the following requirements:

(a) Applicable requirements of 3.5.5.1 at temperatures indicated in Table IV.
(b) At minus 40 degrees Fahrenheit the outer index shall follow the inner index within one-half a graduation width over the full excursion of the range scale in a total elapsed time of 8 seconds for each direction when checked with 18 plus or minus 0.5 volts.

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(c) At plus 150 degrees Fahrenheit using 28 volts, plus or minus 0.5 volts, the ammunition selector set on the APDS-T, M392 Cam, with a load of between 14 and 18 ounce inches applied to the output shaft for the ballistics drive, and with the indexes separated by 3600 meters, the outer index shall stabilize on the inner index and shall meet the applicable requirements of 3.5.4 when the switch is thrown to the "ON" position. Immediately upon completion of the environmental requirements of 3.4, the interior of the computer shall be dried by clean, warm air of between approximately plus 100 degrees to 140 degrees Fahrenheit.

3.4.2.1 Post environmental check.- Each computer which has been subjected to the environmental conditions of 3.4.1 and 3.4.2 shall meet the following requirements after thermal stabilization has been re-established at standard ambient temperature.

(a) The requirements of Table I and 3.5.4.
(b) When checked under power operation at 28 volts, plus or minus 0.5 volt, with 75 to 90 ounce inch load applied, the output shaft for the ballistics drive shall indicate the rotational values of the Table II and within the tolerances 'of Table III for the following ammunition and ranges:

Ammunition	Ranges
APDS-T, M392	500, 1000, 2000 and 3600 meters
HEP-T, M393	500, 1000, 2000 and 3600 meters
HEAT-T, M456 (T384E4)	500, 1000, 2000 and 3600 meters
APERS-T, XM494E3	500, 1000, 2000 and 3600 meters

3.4.3 Vibration.- With the ammunition selector set on the APDS-T, M392 Cam, the outer index set at 3000 meters, the computer (less any shock mounts and brackets, Universal Assembly B8200862 and Output Shaft B8200801) shall be vibrated at an amplitude of 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second for a duration of 5 minutes, plus or minus 15 seconds.

3.4.3.1 Post vibration. - Subsequent to vibration the computer shall not be damaged or show evidence of broken, loose or missing parts and it shall meet the requirements of 3.5 and 3.6.

3.4.4 Sealing.- The computer shall meet the requirements of the rain test of MIL-F-13926.

3.5 Performance.

3.5.1 Output calibration.

3.5.1.1 Zero reference. - Input shaft rotation shall be measured from a precise 1200 meter (zero reference) check point previously determined by the manufacturer and indicated by an index line inscribed on the input shaft in coincidence with an index on the input shaft seal cover plate.

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3.5.1.2 1200 meter checkpoint.- With the computer having a constant load of between 75 and 90 ounce inches applied to the output shaft for ballistics drive, the ammunition selector shall be set on the APDS-T, M392 Cam, With the range correction knob set to zero and with the input shaft set at the 1200 meter checkpoint, actuate the superelevation hand crank to match the outer index with the inner index within plus or minus half a line width by approaching this range first from one direction then from the opposite direction, without overtravel. Note the position of the output shaft for ballistics drive and split the backlash error at this output shaft. This is then the checkpoint for computer output. The position of the superelevation counter shall be established from this checkpoint and shall conform to the requirements of Table I.

TABLE I

Position of Superelevation Counter

Output Shaft Position	117.36° ccw	1200 meter checkpoint 0°	152.64° cw	1R8.64° cw	1R296.64° cw
Tolerance	±3°	±3°	±8°	±8°	±8°
Superelevation Counter Mils	1.25	2.88	5.00	8.00	12.00

3.5.1.3 Outer index and inner index.- Using the APDS-T, M392 Cam only, position the input shaft to correspond with each of the ranges as indicated in Table II and rotate the hand crank until the outer index is in coincidence with the inner index, at which point the rotation of this output shaft shall correspond to the value in Tables II and III. This procedure shall be repeated to check each of the eight other cams at 1200 meters only. All checks shall be made by approaching index from clockwise and counterclockwise direction.

3.5.1.4 Backlash.- Backlash of the output shaft for the superelevator shall not exceed a total of one degree at any point of rotation of the shaft.

3.5.1.5 Outer index and range scale.- Using the APDS-T, M392 Cam only, rotate the hand crank until the outer index is in coincidence with the selected range on the computer range scale for each of the ranges as indicated in Table II, at which point the rotation of the output shaft for balistics drive shall correspond to the values in Tables II and III. All checks shall be made by approaching index from clockwise and counterclockwise direction.

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TABLE II

Computer Accuracy as Determined by Position of Rotation of the Output Shaft Based on 1200 Meter Reference Range for APDS-T, M392 Cam (Drawing No. 8619776).

Range (Meters)	500	1000	1200	1500	2000	3000	3600	4400
Input Shaft Position (±1°)	8R 330.48° cw	1R 308.52° cw	0° Check-Point 0°	2R 98.25° ccw	5R 73.08° ccw	9R 119.88° ccw	11R 68.40 ccw	13R 84.24° ccw
APDS-T, M392	OR 125.28° ccw *1.14	OR 36.72 ccw *2.37	0° *2.88	OR 56.88° cw *3.67	OR 152.64° cw *5.00	OR 357.12° cw *7.84	1R 131.76° cw *9.71	1R 339.12° cw *12.59
HEP-T, M393	OR 273.60° cw *6.68	2R 21.60° cw *13.18	2R 239.04° cw *16.20	3R 244.80° cw *21.28	5R 254.16° cw *31.41	11R 101.52° cw *59.29	15R 284.40° cw *81.83	
HEAT-T, M456 (T384E4)	OR 65.52° ccw *1.97	OR 105.12° cw *4.34	OR 182.88° cw *5.42	OR 312.48° cw *7.22	1R 208.08° cw *10.77	3R 240.48° cw *21.22	5R 228.96° cw *31.06	9R 289.44° cw *51.90
APERS-T XM494E3	1R 109.44° cw *9.4	1R 347.04° cw *12.7	2R 87.84° cw *14.1	2R 231.84° cw *16.1	3R 145.44° cw *19.9	5R 167.04° cw *30.2	6R 354.24° cw *37.8	9R 145.44° cw *49.9

(*) Corresponding nominal Superelevation MIL value.

3.5.1.6 Servo follow-up sensitivity.- With an output shaft load of between 75 and 90 ounce inches, the ammunition selector set for the APOS-T, M392 Cam, the power on at 18 plus or minus 0.5 volts and the automatic follow-up cycle completed at ranges of between 3600 and 4400 meters rotate the input shaft slowly in one direction so that it is possible to count the number of times the contact closes in order to advance the output shaft by approximately 72°. Calculate the average change in superelevation per one make and break of the contacts. Repeat in the opposite direction. The average allowable superelevation increment for each range shall not exceed 10°45′, and the maximum superelevation increment shall not exceed 21°30′ as read at the output shaft.

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3.5.2 Power operation. - With the computer having a constant load of between 75 and 90 ounce inches applied to the output shaft, the rotation bf the output shaft shall correspond to the values indicated in Tables II and III for the ammunition and ranges specified therein The test shall be run by allowing the outer index to approach the inner index from both increasing and decreasing range with power on continuously at 28 plus or minus 0.5 volts. The computer shall also meet the accuracy requirement of Tables II and III for the following ammunition and at ranges indicated when checked at 18 plus or minus 0.5 volts.

Ammunition and Range for 18 Volt Check

Ammunition	Ranges
APDS-T, M392	500, 1000, 2000 and 3600 meters
HEP-T, M393	500, 1000, 2000 and 3600 meters
HEAT-T, M456(T384E4)	500, 1000, 2000 and 3600 meters

All checks shall be made by approaching index from both clockwise and counterclockwise direction.

3.5.3 Output shaft tolerance.- When the output shaft rotation of the computer, as read in degrees is equivalent to the superelevation value in roils as shown in Column "A", the output shaft tolerance shall be within the corresponding value as shown in Column "B" of Table III.

TABLE III

OUTPUT SHAFT TOLERANCE

A Output shaft rotation (Superelevation in mils)	B Degrees	C Mils*
0 up to but less than 6	± 7.20	± 0.10
6 up to but less than 11	± 10.80	± 0.15
11 up to but less than 26	± 14.40	± 0.20
26 up to but less than 31	± 18.00	± 0.25
31 up to but less than 36	± 25.20	± 0.35
36 up to but less than 41	± 28.80	± 0.40
41 up to but less than 61	± 36.00	± 0.50
61 up to but less than 86	± 43.20	± 0.60
86 up to but less than 98.5	± 50.40	± 0.70

*The data contained in Column "C" is informational only.

3.5.4 Stability.- Using 28 plus or minus 0.5 volts with an output shaft load of between 14 and 18 ounce inches, range correction knob set to introcuce minus 15 percent Within one percent, the rotation of the input shaft between minimum range and 1200 meters shall not cause the computer servo to display any self-sustaining or oscillatory operation when

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checked on any of the 4 cams within the computer. A maximum of 8 changes in direction of the servo motor, as a settling out action, shall be considered as satisfactory operation. These 8 changes may be counted as clicks.

3.5.5 Component operation.

3.5.5.1 Operational values.- The operational values shall not exceed the values given in Table IV at temperatures indicated.

TABLE IV

OPERATIONAL VALUES

Control	Operation	Room Temp. to 150°F	Minus 40°F.
Input Shaft	Rotate	16 oz. in.	45 oz. in.
Range Correction Knob	Pull Out	20 lbs.	24 lbs.
	Rotate	34 oz. in.	45 oz. in.
Ammunition Selector	Rotate	60 lb. in.	80 lb. in.
	Translate	10 lbs.	20 lbs.
Handcrank	Hold Engaged	12 lbs.	20 lbs.
	Rotate Engaged	6 lb. in.	20 lb. in.
Reset Button	See Par. 3.6.3	15 lbs.	40 lbs.
Iris Dimmer	Rotate	Mechanically operable	
Interrupter Switches	See Par. 3.6.3	Mechanically and Electrically Operable	

3.5.5.2 Time of response.- At standard ambient temperature (plus 60 to 90 degrees Fahrenheit), with the output shaft load as specified in paragraph 3.5.2 applied, the inner and outer indexes positioned as specified in Table "V, and the ammo selector set on the ADPS-T, M392, the outer index shall stabilize on the inner index within the elapsed time specified, after power has been applied.

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TABLE V

TIME OF RESPONSE

Position of Outer Index	Position of Inner Index	Voltage (plus or minus 0.5.V.)	Total Elapsed Time
4400 meters	500 meters	28 V.	4 seconds
500 meters	4400 meters	28 V.	4 seconds
500 meters	4400 meters	18 V.	8 seconds
4400 meters	500 meters	18 V.	8 seconds

3.5.5.3 Circuit breaker.

3.5.5.3.1 Breaker open. - With the motor at standard temperature, with 28 plus or minus 0.5 volts applied, the two indexes sepearated by at least 200 meters, the output shaft for the ballistics drive locked (immovable), the circuit breaker shall open in 90 seconds or less.

3.5.5.3.2 Breaker closed. - At standard ambient with the load as specified in paragraph 3.5.2 applied to the output shaft for the ballistics drive, the computer APDS-T, M392 Cam, the input shaft shall be oscillated for a period of 45 seconds over a spread of 1000 meters, plus or minus 100 meters between ranges of 1200 to 2800 meters, at a rate of 4 to 4.5 revolutions per second, during which time the circuit breaker shall not open.

3.5.5.4 Hand crank and input shaft travel.

3.5.5.4.1 Hand crank.- The rotation of the superelevation hand crank within the limits of its stops shall cover the excursion from minus 1.5 roils, plus or minus 0.1 mil, to plus 98.5 mils , plus or minus 0.5 mil, as indicated on the superelevation counter when the range correction knob is set at zero.

3.5.5.4.2 Input shaft.- The range input shaft shall rotate through an excursion of 24.0 revolutions plus 0.1 revolution and shall cover the range from 480 to 4575 meters within this excursion, as measured from the 1200 meter checkpoint of the input shaft.

3.5.6 Superelevation cams.- The cams shall be placed in the ballistics computer in the following sequence starting with the HEAT-T, M456 and following with APDS-T, M392, HEP-T, M393 and ADERS-T followed by two spacers. The computer shall be shipped with the same cams with which it qualified.

3.6 Electrical check.

3.6.1 Insulation breakdown. - Insulation breakdown shall be in accordance with MIL-F-13926 except that breakdown voltage of 500 volts RMS shail be substituted for 750 volts RMS.

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3.6.2 Insulation resistance. - Insulation resistance shall be in accordance with MIL-F-13926 except that the resistance value shall not be less than 2 megohms and a dc potential of 500 volts shall be used in place of 800 volts. All lamps shall be removed from sockets prior to tests.

3.6.3 Continuity. - The electrical circuits shall be checked for continuity, and shall conform to the wiring diagram. When the hand crank is engaged the interrupter switch shall open the motor circuit and energize the pilot light. Pressing of the reset button shall cause the motor circuit to be closed and the pilot light to be extinguished. Actuation of the ammo selector handle shall cause the ammunition selector interrupter switch to open the motor circuit. Resetting of the handle shall cause switch to complete the circuit.

3.6.4 Illumination.- The internal lighting device shall provide adequate illumination so that the inner and outer indexes of the range dial and the ammunition index shall be clearly defined and readable when observed in a darkened inclosure.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements, specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 computers plus 3 each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to nsure that it conforms to all the requirements of the contract and submit a record

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of this inspection with the sample, including certificates of conformance for materials.

4.2,2 Contractor testing.- When the contractor i.s responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables VI, VII and VIII herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables VI, VII and VIII shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE VI - CLASSIFICATION OF DEFECTS

CLASS CRITICAL: NONE MAJOR: AQL 1.0% Defective	REQUIREMENTS	TEST PROCEDURE
101. Vibration	3.4.3, 3.4.3.1	4.6.1.5, 4.6.1.5.1
102. Zero reference	3.5.1.1	4.6.2.1
103. 1200 meter checkpoint	3.5.1.2	4.6.2.3
104. Outer index and inner index	3.5.1.3	4.6.2.4
105. Backlash	3.5.1.4	4.6.2.5
106. Outer index and range scale	3.5.1.5	4.6.2.6
107 Servo follow-up sensitivity	3.5.1.6	4.6.2.7
108. Power operation	3.5.2	4.6.2.8
109. Output shaft tolerance	3.5.3	4.6.2.9
110. Stability	3.5.4	4.6.2.10

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TABLE VI - CLASSIFICATION OF DEFECTS Continued

		REQUIREMENT	TEST PROCEDURE
111.	Operational values	3.5.5.1	4.6.3.1
112.	Time of response	3.5.5.2	4.6.3.2
113.	Circuit breaker open	3.5.5.3.1	4.6.3.3.1
114.	Circuit breaker closed	3.5.5.3.2	4.6.3.3.2
115.	Hand crank travel	3.5.5.4.1	4.6.3.4.1
116.	Input shaft travel	3.5.5.4.2	4.6.3.4.2
117.	Superelevation cams	3.5.6	4.6.3.5
118.	Insulation breakdown	3.6.1	4.6.3.6.1
119.	Insulation resistance	3.6.2	4.6.3.6.2
120.	Continuity	3.6.3	4.6.3.6.3
121.	Illumination	3.6.4	4.6.3.6.4

MINOR: NONE

NOTE: The tests in Table VI shall be conducted at a temperature between +60° and 90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One computer shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table VII.

TABLE VII

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Materials	3.1	Applicable drawings - material and general specifications
302.	Fabrication	3.2	Applicable drawings - visual
303.	General specification	3.3	MIL-F-13926 - visual

4.4.2 Environmental. Three computers shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table 1.

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TABLE VIII

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Storage temperature (+160° and -80°F)	3.4.1	4.6.1.1
305.	Operability temperature (+150° and -40°F)	3.4.2, 3.5.5.1	4.6.1.2, 4.6.3.1
306.	Post environmental check (+0° to +90°F)	3.4.2.1	4.6.1.3
307.	Sealing (+60° to +90°F)	3.4.4	4.6.1.4

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. - Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.1 Government furnished inspection equipment. - Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIl-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment

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will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Environmental.

4.6.1.1 Storage temperatures.- Testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.1. The computer shall be exposed to each specified temperature extreme for a minimum of 5 hours. Subsequent to the temperature exposures the computer shall be stabilized in an ambient temperature between +60° and 90°F and subjected to tactile and visual examination to determine compliance with the requirements of 3.4.1. Caution must be exercised during this test to avoid subjecting the computer to thermal shock.

4.6.1.2 Extreme operability temperatures.- These tests shall be conducted in accordance with the temperature conditions in 3.4.2. When thermally stabilized at each of the extreme temperatures specified, the computer shall be tested in accordance with the following:

(a) The controls listed in Table IV shall not exceed the applicable torque and force values at the temperatures specified therein for the operational tests of those controls to determine conformance with 3.4.2a. The equipment required for these tests is listed on Drawing A8565930. The reset button shall be inspected using standard measuring equipment capable of measuring within one-quarter (1/4) pound.

(b) Rotate the input shaft from the clockwise stop to the counterclockwise stop; observe the outer index; it shall follow the inner index in the required time limit at the specified voltage in conformance with 3.4.2b. Repeat the test by rotating the input shaft in the opposite direction. The circuit breaker switch shall be in the "ON" position. This test shall be performed at minus 40 degrees Fahrenheit.

(c) Set the ammunition selector on the cam specified, the range correction knob to zero, and apply the specified torque load to the computer output shaft. Position the computer inner index at 500 meters and the outer index at 4400 meters on the range scale. When the circuit breaker switch is thrown to the "ON" position, the outer index shall "settle out" (cease oscillatory operation) coincident to the inner index within eight (8) changes (clicks) in direction of the computer servo in compliance with 3.4.2c and 3.5.4. The computer shall again meet this requirement when the test is repeated with the outer index positioned at 500 meters and the inner index positioned at 4400 meters on the range scale.

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4.6.1.3 Post environmental temperature checks.- Following temperature environmental tests, the computer shall be thermally stabilized at room temperature (60° to 90°F) and tested for the following to determine compliance with 3.4.2.1:

(a) The tests of 4.6.2.3 and 4.6.2.9

(b) With the required load applied to the output shaft and at the specified voltage, the computer shall be tested in accordance with 4.6.2.8.

4.6.1.4 Sealing (water resistance). - This test shall be conducted in accordance with Procedure I of MIL-F-13926 to determine compliance with 3.4.4. The computer shall show no evidence of water penetration as a result of this test. The receptacle shall be covered with waterproof tape during this test. The computers that are opened for environmental drying and examination for water penetration shall be resealed to meet the requirements of the drawings prior to submission for final acceptance.

4.6.1.5 Vibration. - Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.4 shall be utilized for the vibration test. Unless otherwise specified, the test equipment shall be in conformance with the “Test Facilities” requirements of MIL-F-13926. Position the computer on the vibration table in accordance with the requirements of 3.4.3. The computer shall be vibrated in a vertical plane for a period of 5 minutes ± 15 seconds.

4.6.1.5.1 Post vibration, physical failure. - Following vibration, the computer shall be subjected to a visual and tactile examination for damage such as loose, bent or broken parts, et. to determine compliance with 3.4.3.1. The computer shall then be subjected to all the following tests of this specification.

4.6.2 Output calibration.

4.6.2.1 Zero reference.- The 1200 meter checkpoint as determined previously by the manufacturer shall be visually checked for compliance with 3.5.1.1. It shall be indicated by an index line inscribed on the input shaft in coincidence with an index on the input shaft seal plate cover.

4.6.2.2 The following tests shall be conducted using Test Fixture A8565930 and in accordance with the operating instructions specified on Drawing D8258567 using appropriate metric scales. The computer input shaft and the fixture input mechanism shall be oriented to the 1200 meter checkpoint.

4.6.2.3 1200 meter checkpoint.- Set the ammunition selector on the cam specified and the range correction knob on zero. When the mil value specified in Table I is indicated on the computer superelevation mil

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counter, the rotational position of the output shaft for the ballistics drive shall not exceed the tolerance specified in Table I for compliance with 3.5.1.2.

4.6.2.4 Outer index and inner index. - Set the ammunition selector on the cam specified and the range correction knob on zero. When the rotational position of the computer input shaft corresponds to the value for each of the ranges indicated in Table II, and the computer outer and inner indexes are coincident, the rotational position of the ballistics drive output shaft shall not exceed the tolerances as specified in Table III. This test shall be repeated to check each cam at 1200 meters only for compliance with 3.5.1.3.

4.6.2.5 Backlash. - Backlash of the computer superelevator output shaft shall not the tolerance specified in 3.5.1.4 when measured at four (4) points of shaft rotation. The four (4) points of measure shall be approximately ninety degrees (90°) apart.

4.6.2.6 Outer index and range scale. - Set the ammunition selector on the cam specified and the range correction knob on zero. When the computer outer index is coincident to the selector range mark on the computer range scale, the rotational position of the output shaft for the ballistics drive shall correspond to the values of Table II and not exceed the tolerances of Table III for compliance with 3.5.1.5.

4.6.2.7 Servo follow-up sensitivity.- Set the ammunition selector on the cam specified, the range correction knob on zero, and the inner and outer index to a range between 3600 meters and 4400 meters. With the computer in power operation, the average and maximum recorded increment change (closing of computer servo contacts) resulting from a total rotation of the output shaft 72° clockwise and 72° counterclockwise shall not exceed the tolerance specified in 3.5.1.6. Superelevation increment changes shall be measured at the computer output shaft for the ballistics drive.

4.6.2.8 Power operation.- This test shall be conducted as specified on Drawing D8258567 with the range correction knob set to zero. The tests in 4.6.2.8.1 and 4.6.2.8.2 shall be conducted to determine compliance with 3.5.2.

4.6.2.8.1 Input at 28 volts.- With the input voltage set at 28 plus or minus 0.5 volts and the circuit breaker switch in the “ON” position, the accuracy of each cam shall be tested at the range listed in Table II for computer output shaft rotation. When the input shaft is rotated to a value indicated in Table II for a given range, the rotational position of the output shaft shall correspond to the value for that range in Table II within the tolerance specified in Table III to determine compliance with 3.5.2.

4.6.2.8.2 Input at 18 volts. - With the input voltage set at 18 plus or minus 0.5 volts, the computer shall meet the accuracy requirements of Tables II and III for the cams and ranges specified to determine compliance with 3.5.2.

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4.6.2.9 Output shaft tolerance.- Rotation of the output shaft shall be determined by representative roils in superelevation (see Column A of Table III) and shaft rotation measurement in degrees shall be within the tolerance specified in Column C of Table III for conformance with 3.5.3.

4.6.2.10 Stability. - Prepare the computer for this test in accordance with Drawing D8258567, set the range correction knob to minus fifteen percent (-15%) within one percent (1%), and adjust the input voltage to 28 plus or minus 0,5 volts. Rotate the computer index from minimum range to 1500 meters; throw the circuit breaker switch to "ON" position to energize the servo motor. The outer index shall "settle out" (Cease oscillatory operations) coincident to the inner index within eight (8) changes (clicks) in direction of the computer servo. The inner index shall then be rotated to four (4) different ranges between 1500 meters and minimum range, and the outer index again shall "settle out" coincident to the inner index within eight (8) changes in direction of the servo. Set the range correction knob to plus fifteen percent (+15%) within one percent (1%), and repeat the aformentioned test. All cams installed in or accompanying the computer shall be subjected to this test to determine compliance with 3.5.4.

4.6.3 Component operation.

4.6.3.1 Operational values.- Operating tests shall not exceed the values listed in Table IV at the temperatures indicated to determine compliance with 3.5.5.1.

4.6.3.2 Time of response.- Perform this test on Test Fixture D8585200 utilizing the operating instructions on Drawing D7660281. Set the ammunition selector on the cam specified and the range correction knob on zero. When the circuit breaker switch is thrown to the "ON" position, and the servo motor is energized with the load in 3.5.2 applied to the output shaft, the computer outer index shall stabilize on the inner index within the time cycle specified in Table V for compliance with 3.5.5.2.

4.6.3.3 Circuit breakers.- Perform the tests in 4.6.3.3.1 and 4.6.3.3.2 on Test Fixture D8585200 utilizing the operating instructions on Drawing D7660281.

4.6.3.3.1 Circuit breaker, open.- The inner and outer index shall be separated by the specified distance, and the output shaft for the ballistics drive immobilized. When the circuit breaker switch is thrown to the "ON" position, and the servo motor is energized, the circuit breaker shall open (switch move to "OFF" position) within the time cycle specified in 3.5.5.3.1.

4.6.3.3.2 Circuit breaker, closed.- Set the ammunition selector on the cam specified, the range correction knob on zero, and the inner and outer index coincident on 2000 meters of range. The computer input shaft

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shall be oscillated as specified in 3.5.5.3.2 with the circuit breaker switch in the "ON" position and the servo motor energized. The circuit breaker shall not open (switch move to "OFF" position) as a result of this oscillation.

4.6.3.4 Hand crank and input shaft travel.- Perform the tests in 4.6.3.4.1, 4.6.3.4.2 and 4.6.3.5 on Test Fixture A8565930 utilizing the operating-instructions on Drawing D8258567.

4.6.3.4.1 Hand crank travel.- The test for hand crank travel shall be conducted by manually rotating the hand crank and observing the super-elevation counter to determine the excursion specified in 3.5.5.4.1.

4.6.3.4.2 Input shaft travel.- The test for input shaft travel shall be conducted by manually rotating the range input shaft through the ex-cursion specified in revolutions in 3.5.5.4.2 to determine if full range coverage (in meters) is in compliance with 3.5.5.4.2 as measured from the 1500 meter checkpoint of the input shaft.

4.6.3.5 Superelevation cams.- The inspection sequence and arrangement of cams in the computer shall be as specified in 3.5.6 by means of visual examination.

4.6.3.6 Electrical checks. - The following tests shall be ocnducted at any time subsequent to vibration. Before testing, remove lamp bulbs and disconnect capacitor from computer circuit.

4.6.3.6.1 Insulation breakdown.- Procedure I in this test shall be conducted in accordance with MIL-F-13926 for insulation breakdown to meet the requirements of 3.6.1.

4.6.3.6.2 Insulation resistance.- This test shall be conducted in accordance with MIL-F-13926 for insulation resistance, except the values in 3.6.2 shall apply.

4.6.3.6.3 Continuity. - A continuity test of the computer circuits shall be made using standard measuring equipment to determine conformance with the wiring diagram. Actuation of the hand crank, reset button, and ammunition selector handle shall cause the computer to function.

4.6.3.6.4 Illumination. - The illuminated inner and outer indexes of the range dial and ammunition index of the computer shall meet the requirements When observed in a darkened enclosure with the unaided eye.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing, and marking shall be accordance with the applicable packaging data sheet listed in section 2. The level of protection shall be as specified in the procure-ment document.

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6. NOTES

6.1 Intended use. - The Ballistics Computers are electromechanical components of the primary fire control system for 105 MM Gun Tank: M60A1. The ballistics computers manually or electrically determine and relay to the sighting system the amount of gun superelevation required for specific target range, and ammunition type.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of preservation, packaging and packing.
c. Applicable stock number.
d. Applicable packaging data sheet (see 5.1).
e. Requirement for First Article testing.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1220-A250

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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